Benchmarking Protein–Protein Interface Prediction on Experimental versus AlphaFold-Predicted Structures

Armeen Shasti-Nazem*
University of Washington, Seattle, WA, USA
aashasti@uw.edu

Draft of October 1, 2026

Abstract

PeSTo is a geometric deep-learning model that predicts protein–protein interface residues from atomic coordinates, trained only on experimentally determined structures. Because most protein structures available today are AlphaFold predictions, we ask whether PeSTo's accuracy transfers to AlphaFold input. Starting from PDB complexes released after AlphaFold2's training cutoff and filtered to under 30% sequence identity to PeSTo's own training, validation, and test sequences, we assembled a primary benchmark set of 566 protein chains with both an experimental and an AlphaFold structure, a validated residue mapping, and interface labels, and ran PeSTo on each chain's isolated experimental structure and on the matched AlphaFold model trimmed to the same residue range. Across 554 chains with defined metrics, PeSTo's per-chain area under the precision–recall curve (AUPR) was higher on experimental input by a median of 0.027 (95% CI [0.019, 0.035], Wilcoxon $p < 10^{-10}$), with the same direction in every subgroup of four pre-specified descriptive splits and under an alternative buried-surface-area interface definition. The gap was far from uniform: within AlphaFold's own confidence bands, the pooled AUPR gap shrank from 0.254 in the lowest-confidence residues to 0.039 in the highest-confidence residues, and a per-residue regression associated both AlphaFold confidence (pLDDT) and local structural divergence with how much each residue's prediction changed between inputs, independently of one another and of solvent exposure (and, in a post-hoc refit, with AlphaFold-input predictions lying further from the true label). Chain length was not associated with the size of the drop. These results describe a modest but consistent, confidence-linked accuracy cost when applying PeSTo to AlphaFold structures. The per-residue cost is largest in lower-confidence regions, where interface residues are over-represented, although most interface residues (57%) lie in high-confidence (pLDDT ≥ 90) regions. All reported relationships are associations, not demonstrated causes.

1 Introduction

Protein–protein interactions underlie nearly all cellular processes, from signal transduction to immune recognition, and the specific residues that form a binding interface are relevant to understanding disease mutations and to structure-based drug design [1]. Determining an interface experimentally, by solving the structure of a complex, remains slow relative to the scale of the problem: the Protein Data Bank (PDB) holds on the order of 2×10^5 structures, while the AlphaFold Protein Structure Database now provides predicted structures for over 2×10^8 sequences [2, 3]. Several geometric deep-learning methods now predict interface residues directly from a single protein's three-dimensional coordinates, without requiring a solved complex: PeSTo represents atoms by element and position alone [5], and ScanNet extracts learned geometric and biochemical descriptors [6]. These models are trained on experimentally determined structures, and applying them to AlphaFold predictions implicitly assumes the predicted geometry is close enough to the true structure that the model's learned decision boundary still applies.

That assumption is worth checking directly. AlphaFold reports a per-residue confidence score, pLDDT. Its documentation states that a pLDDT above 70 usually corresponds to a correct backbone prediction, while regions below 50 are often flexible or intrinsically disordered [4]. Interface residues are over-represented among low-confidence residues: in the dataset analyzed here, the interface base rate is 0.40 among residues with pLDDT below 50 versus 0.19 among those at or above 90 (Table 1). Even confidently predicted regions deviate measurably from experimental structures: Terwilliger et al. report median inter-atomic distance deviations, within regions of

*Student, University of Washington, Seattle, WA, USA. This project was conducted independently, outside of any course, laboratory, or supervised research program, and does not represent the University of Washington.

pLDDT above 70, of about 0.1 Å for nearby atom pairs rising to about 0.7 Å for pairs roughly 50 Å apart [7]. Yuan et al. reported that PeSTo's median per-protein AUPR on PeSTo's own protein–protein test set fell from 0.797 with experimental input (the value reported by PeSTo's authors) to 0.691 with ESMFold-predicted input [8, 9]. ESMFold is a protein-language-model-based structure predictor distinct from AlphaFold, so this is a substantial drop but not a direct measurement of AlphaFold specifically, whose error characteristics differ from ESMFold's.

This paper reports a matched, AlphaFold-specific benchmark of PeSTo together with an error analysis of where any accuracy gap occurs. We ask two questions: (1) how much does PeSTo's interface-prediction accuracy change when the input is an AlphaFold model of a chain rather than that chain's own experimental structure, holding the chain and the ground-truth interface fixed; and (2) is that change associated with AlphaFold's own confidence score and with the local geometric divergence between the two structures. Both analyses followed written analysis plans fixed before they were run on real predictions (Section 2.4 describes exactly what was fixed when). PeSTo's authors themselves applied it to 7,464 high-quality human AlphaFold DB models, selected by pLDDT and PAE, and reported that agreement with predicted yeast complexes improved in higher-quality model regions [5]. This study extends that work and Yuan et al.'s ESMFold benchmark to a controlled, AlphaFold-specific comparison on matched experimental structures, with a paired error analysis; it is not the first study to raise this general concern.

2 Methods

2.1 Dataset construction

Candidate protein chains were drawn from X-ray crystal structures in the PDB released after 2018-04-30, AlphaFold2's own training-data cutoff, so that the experimental structures used here were not part of AlphaFold2's training data (see Limitations regarding the templates used to generate the AlphaFold DB models). Candidates were restricted to entries at or better than 2.5 Å resolution with no nucleic-acid polymer entities, complexes of at most 10 distinct protein entities (excluding assemblies too large for a single well-defined pairwise interface, e.g. viral capsids), and chains whose polymer entity sequence is at least 40 residues long (the entity's full expressed/construct sequence, not the count of residues actually resolved in the deposited coordinates, which is often substantially smaller). Each chain's UniProt accession from the RCSB Data API was cross-checked against the SIFTS [13] bulk chain-to-UniProt mapping (release header: 2026/09/13 - 09:34; PDB: 37.26; UniProt: 2026.04). This search returned 8,808 PDB entries and 22,887 candidate protein chains (Fig. 1). Redundant chains were then collapsed with MMseqs2 [12] (version `d401e78`) `easy-cluster` at `--min-seq-id` 30% and `-c` 80% with `--cov-mode 0` (the alignment must cover at least 80% of both sequences), all other parameters at their defaults (greedy set-cover clustering, sensitivity 4, E-value 10^{-3}), to 3,009 non-redundant representatives; each cluster's representative is its best-resolution member, then its longest, then the first by PDB and chain identifier.

Because PeSTo was trained on essentially all biological assemblies in the PDB, filtering by exact PDB identifier is not sufficient: a chain released after 2018 can still be a close sequence homolog of something PeSTo has already seen under a different identifier. Each representative was therefore searched with MMseqs2 `easy-search` against the union of PeSTo's published training, validation, and test sequence sets (same version; `--min-seq-id` 30%, `-c` 80%, `--cov-mode 0`, default sensitivity 5.7 and E-value 10^{-3}; identity is MMseqs2's identical positions over alignment length). 77% of representatives had a hit at or above 30% identity and were treated as leakage. The leakage-free candidate set carried forward into the rest of the pipeline is the 696 representatives below this threshold; not all of them survive the remaining processing steps below, which is why this paper distinguishes the leakage-free candidate set (696) from the smaller primary benchmark set that is actually analyzed in Section 3 (566, defined below).

For each leakage-free representative, the PDBe "updated" mmCIF file (downloaded 2026-09-16) and the corresponding AlphaFold DB model (queried through the AlphaFold DB prediction API and downloaded 2026-09-16) were retrieved; 609 representatives had both. The rest had no AlphaFold DB entry for their UniProt accession (66), were split into fragments in AlphaFold DB (13), or had an AlphaFold DB sequence that did not match the UniProt sequence (8). Every AlphaFold model in the primary benchmark set is AlphaFold DB model version v6, generated by the "AlphaFold Monomer v2.0 pipeline" (model creation date 2022-06-01 to 2025-08-01 in the API metadata). AlphaFold DB also serves third-party community models under the same per-accession query as official AlphaFold2 predictions; these were identified from the API metadata's provider field and excluded (14 representatives, provider VR3D, e.g. ColabFold models), since they do not carry AlphaFold2's training-cutoff guarantee.

Residue numbering between the experimental structure and the AlphaFold model (which uses UniProt numbering)

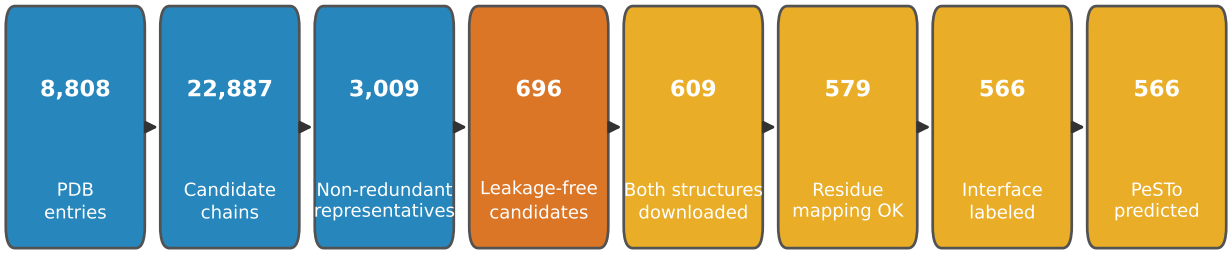


Figure 1: Dataset construction: candidate PDB chains are reduced by redundancy clustering, sequence-level leakage filtering against PeSTo's own training/validation/test data, structure availability, residue-mapping validation, and interface labeling, to the 566-chain primary benchmark set. Counts from "Leakage-free candidates" onward are for the leakage-filtered chains only.

was mapped residue by residue using the SIFTS cross-references embedded in each PDBe updated mmCIF file, falling back to a pairwise sequence alignment when SIFTS was unavailable or failed a $\geq$90% residue-identity validation check. Chains were excluded if neither method passed validation (8) or if fewer than 80% of the chain's observed residues mapped to a position within the AlphaFold model (8). 579 representatives obtained a validated mapping. Of the 566 chains in the final primary set, 540 used the SIFTS mapping and 26 the fallback alignment; 97 contain at least one residue that differs from the AlphaFold model's sequence (minimum identity 92%), typically engineered point mutations.

Ground-truth interface residues were computed from coordinates in the depositor-annotated biological assembly – not the raw crystallographic asymmetric unit, which is dominated by crystal-packing contacts unrelated to biological interfaces, and not the PISA assembly-annotation service [10], which has no practical batch interface for a set this size. For each chain, the first author-determined assembly containing it was used (otherwise the first listed assembly containing it; the asymmetric unit itself only for entries that define no assembly). A residue is an interface residue if any of its heavy atoms lies within 5 Å of any heavy atom of a standard or modified amino-acid residue on another chain of that assembly, including symmetry-generated copies; ligands, ions, waters, and nucleic acids are not partners. Alternate conformations were resolved per atom by highest occupancy, and hydrogens were ignored throughout. A second, buried-surface-area–based definition (loss of at least 1 Å^2 of solvent-accessible surface area when the protein partners are added) was retained as a robustness check on the labeling choice; SASA was computed with Biotite's Shrake–Rupley implementation [14, 15] (version 1.6.0, ProtOr radii [16], probe radius 1.4 Å, 1,000 points per atom). Chains were excluded at this step if their assembly contained no protein partner (12) or no residue met the interface criterion (1), 13 in total. The remaining 566 representatives form the primary benchmark set analyzed in Section 3; a full list is provided as Supplementary Table S1 (Code and Data Availability).

2.2 Inference

The pretrained PeSTo model (repository commit `ba651aa`, checkpoint `i_v4_1_2021-09-07_11-21`, which takes atomic elements and coordinates as input) was run on each of the 566 primary-set chains in two input forms. The experimental input (*exp*) is the chain's own heavy atoms from the PDBe mmCIF file – every observed polymer residue of the chain, with other chains, waters, ligands, and ions removed, alternate conformations resolved as above, and modified residues (e.g. selenomethionine) kept as deposited; missing atoms were not rebuilt. The AlphaFold input (*af_trimmed*) is the AlphaFold model's heavy atoms for every residue between the first and last mapped UniProt position, inclusive; it therefore also contains residues inside that range that are unresolved in the experimental structure, while the experimental input contains any observed residues that do not map to UniProt (e.g. expression tags). Both runs used the chain in isolation, with no partner structure present, matching how an interface predictor would be applied to a single AlphaFold model in practice. We used PeSTo's protein–protein interface output (channel 0, sigmoid-transformed). All metrics are computed only on the residues present in both inputs and mapped between them, so each chain is scored on an identical residue set and against identical labels for both inputs.

2.3 Derived structural features

Relative solvent accessibility (RSA) is the residue's SASA in the isolated experimental chain (its bound conformation with partners removed) divided by the theoretical maximum of Tien et al. [17]; "surface" residues are those with RSA ≥ 0.25. Secondary structure (helix, strand, coil) was assigned from the experimental Cα trace by P-SEA [18] as implemented in Biotite (DSSP was not available in the computing environment). pLDDT was read from the AlphaFold model's B-factor column. A residue's local Cα RMSD uses the mapped residues whose experimental Cα lies within 10 Å of its own (itself included; at least 3 required), with a separate Kabsch superposition of that neighborhood's Cα atoms and the RMSD taken over the same neighborhood, so that structural motion elsewhere in a multi-domain chain is not mistaken for local error at a given residue.

A chain is *geometrically flagged* if, after one global Kabsch superposition of all its mapped Cα atoms onto the AlphaFold model's, at least 5 consecutive mapped residues (in UniProt order) are more than 10 Å apart. This check was built to catch residue mappings that pass sequence validation but do not correspond in 3D; in practice it also flags genuine large-scale conformational differences, such as domain rearrangements.

2.4 Pre-specified analysis plans

Sections 3.1 and 3.2 follow analysis plans written into the project's design document before each analysis was run on real predictions. These plans are pre-specified, not publicly pre-registered, and the record of their timing is limited in one specific way. The original project plan (committed 2026-09-16, `ac34866`, before any data were downloaded) already specified the paired Wilcoxon test with a bootstrap CI on per-chain AUPR, pLDDT bands at 50, 70, 90, local RMSD, RSA, and secondary structure as covariates, and flagging of predictors with VIF above 5. The detailed plans (exact endpoints, strata, thresholds, and regression specification) were written later and committed to version control in the same commits as the corresponding results (`f289b6b` and `8db13d4`). That they preceded the results therefore rests on the author's own records, not on an independent timestamp. The error-analysis plan was written after the benchmark results were known. It also changed the original plan's regression outcome (a binary prediction-error indicator, modeled by logistic regression) to the continuous prediction shift described below, before any error-analysis feature was computed.

Post-hoc analyses added after review. Five further analyses were chosen after all of the above results were known, in response to peer review (code: `src/analysis/posthoc_review.py`). They are exploratory, are labeled "post-hoc" wherever they appear, and do not replace any pre-specified result. (1) *Template window:* each chain's PDB release date was compared with its AlphaFold DB model's creation date, and the primary endpoint repeated on the chains released on or after that date, which cannot have served as templates for their own models. (2) *Signed error:* the regression design above was refitted with $|p_{\mathrm{AF}} - y| - |p_{\mathrm{exp}} - y|$ as the outcome, where y is the true label (positive when the AlphaFold-input prediction is further from the truth), and R^2 is reported for both outcomes. (3) *Paired F1 bootstrap* for Section 3.3: the change in F1 between the smallest and largest pLDDT cutoff, and the change in the experimental-vs.-AlphaFold F1 gap, with both cutoffs evaluated on the same resampled chains (2,000 resamples). (4) *Pooled vs. per-chain gap:* pooled AUPR recomputed after removing the 10% of chains with the largest per-chain drops, and after replacing each chain's scores by their within-chain percentile rank, which removes differences in score calibration between chains. (5) *Flag audit:* mapping method and identity of geometrically flagged vs. unflagged chains, and a per-chain listing of residues with local RMSD above 10 Å.

Benchmark (Section 3.1). Primary and secondary endpoints are the paired per-chain difference (experimental minus AlphaFold-trimmed) in AUPR and ROC-AUC respectively, restricted to chains where both metrics are defined (excluding chains whose scored residues are single-class). Each was tested with a two-sided Wilcoxon signed-rank test ($\alpha = 0.05$) and a 95% percentile bootstrap CI on the median difference (10,000 resamples of chains, not residues, with replacement, since residues within one chain are not independent observations; fixed seed 0). Pooled-residue AUPR and ROC-AUC were reported for each input. Four binary splits were specified as descriptive strata, reported side by side with n, not as additional hypothesis tests, and not corrected for multiple comparisons: homomeric vs. heteromeric partner, interface fraction below vs. at or above 0.5, geometrically flagged vs. not, and all residues vs. surface residues only. A robustness check repeats the primary endpoint with the buried-surface-area label.

Error analysis (Section 3.2). pLDDT confidence bands are fixed at 50, 70, 90 (AlphaFold's own convention). Pooled AUPR/ROC-AUC are computed for both inputs within each band, with chain-resampled 95% percentile bootstrap CIs (2,000 resamples per band, fewer than the benchmark's 10,000 because each draw here recomputes a full pooled metric over tens of thousands of residues rather than a cheap scalar; fixed seed 0). A per-residue ordinary-least-squares regression relates the absolute prediction shift, $|p_{\mathrm{AF}} - p_{\mathrm{exp}}|$, to standardized pLDDT,

standardized local Cα RMSD, standardized RSA, and secondary-structure class (coil as reference). Standard errors are cluster-robust (CRV1), clustered by chain, and variance inflation factors (VIF) are reported for each predictor (flagged above 5, since pLDDT and local RMSD both reflect AlphaFold-vs-experimental divergence and were expected to correlate). Descriptive analyses of chain length and of the geometrically flagged chains were specified as associations to be reported, not confirmatory tests. In the results, p-values below 10^{-10} are reported only as that bound.

Which chains enter which analysis. Per-chain analyses (the paired endpoints, the descriptive strata, and the chain-length and flagged-chain comparisons) use the 554 chains for which per-chain metrics are defined. Residue-level analyses (pooled metrics, pLDDT bands, the shift regression, and Section 3.3) use every scored residue of all 566 chains, including the 136 residues (0.1% of the pooled total) of the 12 single-class chains described in Section 3.1.

3 Results

3.1 Benchmark: experimental versus AlphaFold-trimmed input

Of the 566 primary-set chains, 12 (2.1%) were excluded from every per-chain comparison because every scored residue in the chain was labeled interface (these chains have only 6–26 scored residues), leaving no non-interface residues against which to compute ROC-AUC or AUPR; the remaining 554 chains form the primary comparison.

PeSTo was more accurate on experimental input than on the matched AlphaFold-trimmed input: the median paired AUPR difference was 0.027 (95% CI [0.019, 0.035], Wilcoxon $p < 10^{-10}$), and the paired ROC-AUC difference showed the same pattern (median 0.027, CI [0.018, 0.037], $p < 10^{-10}$). The direction was not universal: AlphaFold input gave the higher AUPR for 178 of 554 chains (32%), experimental input for 375, with 1 tie. Pooling all 95,118 scored residues (interface base rate 0.223) rather than comparing per chain gives AUPR 0.739 on experimental input versus 0.669 on AlphaFold input (pooled ROC-AUC 0.882 vs. 0.849). Both inputs are far above the base rate, so both are informative, but the pooled AUPR gap (0.070) is noticeably larger than the per-chain median gap; Section 3.2 returns to this. Replacing the distance-based interface label with the buried-surface-area label gives essentially the same result (median 0.026, CI [0.020, 0.034], $p < 10^{-10}$), so the finding is not an artifact of the distance-cutoff label definition.

A post-hoc check addressed whether AlphaFold DB could have used a chain's own experimental structure as a template. 334 of the 566 chains were released before their AlphaFold model was generated, so they could in principle have been templates for it. This is an upper bound, because the template database predates the model. The remaining 232 chains cannot have been. Among those (229 with defined metrics), the median paired AUPR difference was 0.032 (95% CI [0.020, 0.047], $p = 6.6 \times 10^{-8}$), versus 0.022 (CI [0.016, 0.033], $p = 1.6 \times 10^{-10}$) among the 325 that could have been. The gap is therefore present without any possibility of self-templating. Its point estimate is larger in that subgroup, consistent with self-templating narrowing the gap where it was possible, but the two subgroups were not formally compared.

Across the four pre-specified descriptive splits (Fig. 3), every subgroup showed the same direction of effect, with medians from 0.015 ($n = 120$, Interface fraction ≥ 0.5) to 0.059 ($n = 92$, Geometrically flagged), and no subgroup's bootstrap CI included zero. (The all-residues arm of the residue-scope split is the primary endpoint itself, so 7 distinct subgroups are shown alongside it.) As pre-specified, these are descriptive comparisons and are not interpreted as further hypothesis tests. The gap is therefore not confined to one kind of chain, though its size varies noticeably. It is largest among the minority of geometrically flagged chains, those whose AlphaFold model and experimental structure disagree over a long contiguous stretch after superposition (Section 2.3).

3.2 Error analysis: what the gap is associated with

Stratifying every scored residue by AlphaFold's own pLDDT confidence band (Fig. 4, Table 1) shows the AUPR gap shrinking monotonically as confidence increases, from 0.254 in the lowest band to 0.039 in the highest – roughly a 6.4-fold reduction (computed from unrounded values) – with the same pattern in ROC-AUC. Every band cleared the minimum-residue reporting threshold with no undefined bootstrap draws.

A per-residue regression of the absolute prediction shift on standardized pLDDT, local RMSD, RSA, and secondary structure (Fig. 5, Table 2; $n = 95,117$ of the 95,118 pooled residues,[1] 566 chains as clusters, cluster-robust standard

[1] 1 residue was excluded for a missing RSA value: a single non-standard N-terminal residue whose chemical component has no reference maximum accessible surface area in the table of Tien et al. [17]. It was logged during interface labeling and is not otherwise

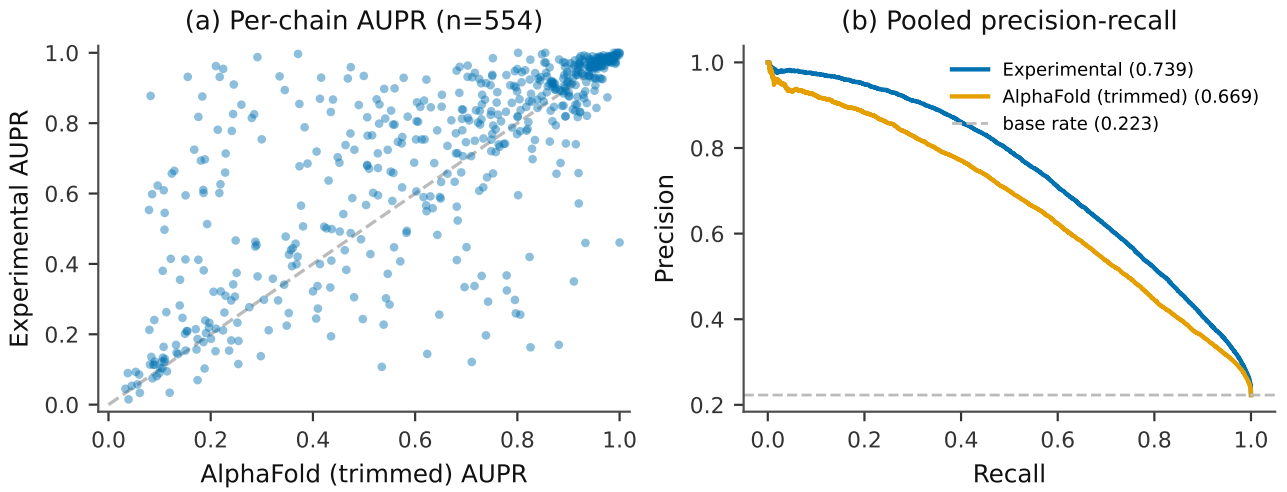


Figure 2: (a) Per-chain AUPR, experimental vs. AlphaFold-trimmed input; most chains sit near the identity line with a tilt toward experimental input, and a right-skewed tail where AlphaFold input costs noticeably more accuracy. (b) Precision–recall curves pooling every scored residue in the primary set, for each input, against the pooled interface base rate.

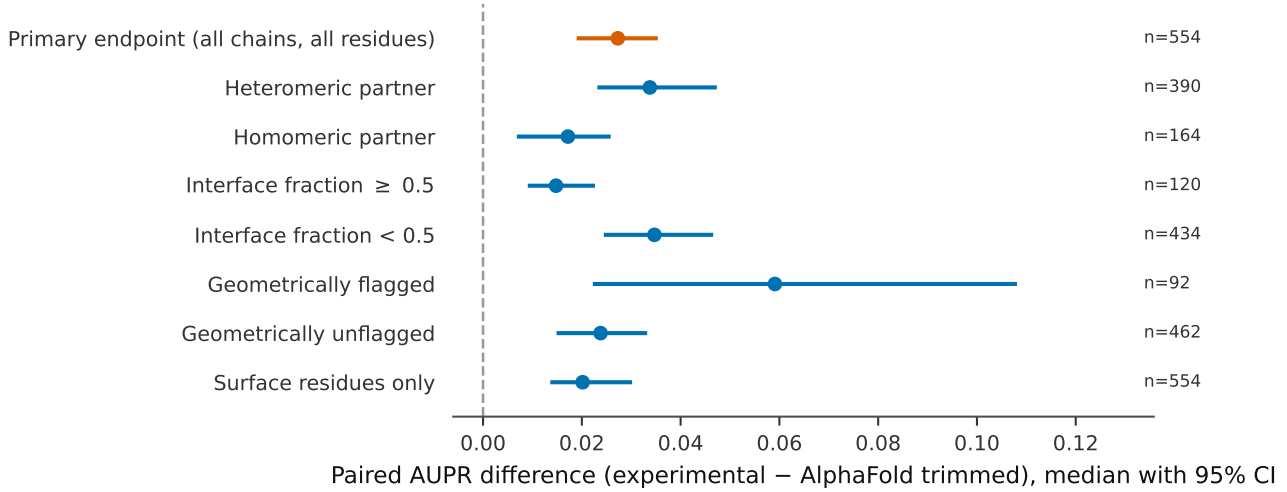


Figure 3: Paired AUPR difference (median, 95% chain-resampled bootstrap CI) for the primary endpoint and the subgroups of the four pre-specified descriptive splits. The all-residues subgroup is identical to the primary endpoint and is shown once. All intervals lie on the same side of zero.

errors) finds pLDDT (coefficient -0.0268, $p < 10^{-10}$) and local RMSD (coefficient 0.0268, $p = 1.4 \times 10^{-9}$) each independently associated with the size of the shift in the expected direction – higher confidence is associated with a smaller shift, larger local structural divergence with a larger one – after controlling for the other and for RSA. RSA had the strongest association of any predictor (coefficient 0.0254, $t = 16.0$, $p < 10^{-10}$, versus $|t| = 8.1$ for pLDDT and 6.1 for local RMSD): predictions for more exposed residues change more between inputs. No predictor's variance inflation factor exceeded 5 (maximum 1.64): although both are proxies for AlphaFold-vs-experimental divergence, pLDDT and local RMSD are only mildly collinear here and each retains independent explanatory power. Binned by local-RMSD decile, the mean absolute prediction shift changes little across the lowest deciles and rises over the upper ones, most steeply in the top decile (Fig. 5a). Nearly all residues with extreme local RMSD (798 of the 805 residues above 10 Å, in 28 of 29 chains) belong to geometrically flagged chains, 16 of which were mapped by SIFTS with 100% sequence identity to the AlphaFold model. This makes residue-register errors an unlikely explanation for most of them, though it does not rule them out for individual chains.

The shift measures how much a prediction moves between inputs, not whether it moves toward or away from the true label, and the pre-specified regression explains only a small part of its variance ($R^2 = 0.093$). A post-hoc refit

informative for this regression.

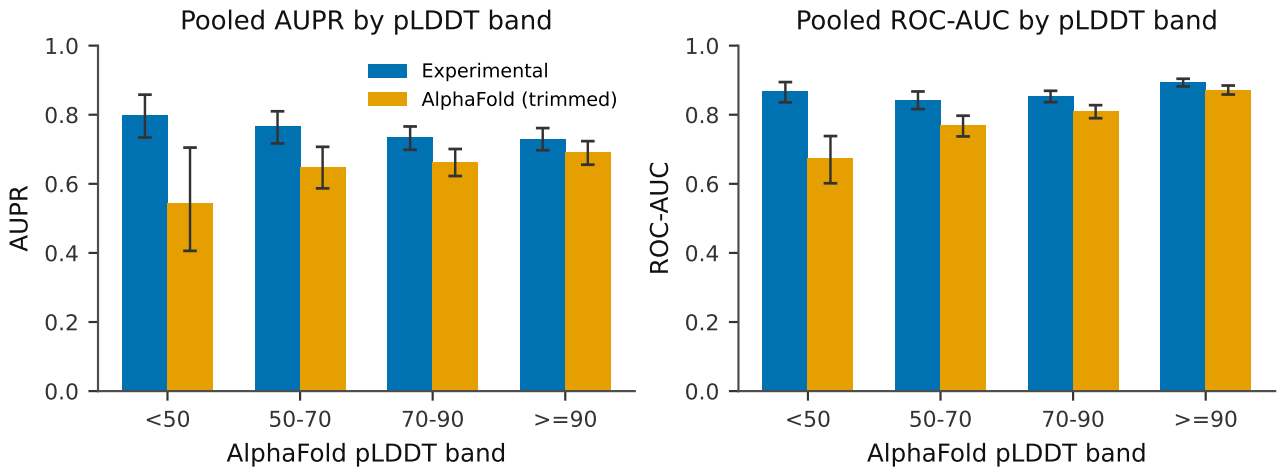


Figure 4: Pooled AUPR and ROC-AUC by AlphaFold pLDDT confidence band, both inputs, with 95% chain-resampled bootstrap CIs. The experimental-vs.-AlphaFold gap is largest in AlphaFold's least confident residues and smallest in its most confident ones.

Band	n res.	n ch.	Base rate	Input	ROC-AUC (95% CI)	AUPR (95% CI)
<50	2,828	216	0.400	Experimental	0.868 [0.836, 0.894]	0.798 [0.734, 0.858]
				AlphaFold trimmed	0.673 [0.602, 0.738]	0.543 [0.406, 0.705]
50-70	5,735	445	0.332	Experimental	0.843 [0.816, 0.867]	0.766 [0.717, 0.810]
				AlphaFold trimmed	0.768 [0.737, 0.797]	0.649 [0.587, 0.707]
70-90	23,762	535	0.256	Experimental	0.853 [0.836, 0.869]	0.734 [0.699, 0.766]
				AlphaFold trimmed	0.809 [0.790, 0.828]	0.663 [0.623, 0.701]
>=90	62,793	493	0.192	Experimental	0.893 [0.882, 0.904]	0.730 [0.697, 0.761]
				AlphaFold trimmed	0.871 [0.858, 0.885]	0.690 [0.655, 0.724]

Table 1: Pooled ROC-AUC and AUPR by pLDDT band, both inputs, with 95% chain-resampled bootstrap CIs.

with the signed error $|p_{\mathrm{AF}} - y| - |p_{\mathrm{exp}} - y|$ as the outcome (Table 3) separates the two. On average AlphaFold-input predictions are slightly further from the true label (mean 0.018, CI [0.013, 0.024]; further for 53% of residues). Lower pLDDT (coefficient -0.0145, $p = 7.1 \times 10^{-5}$) and higher local RMSD (0.0233, $p = 8.2 \times 10^{-5}$) are both associated with AlphaFold input being further from the truth, not merely with a change. Solvent exposure, the strongest correlate of the shift, is not associated with the signed error (-0.0012, $p = 0.473$). Exposed residues' predictions move more between inputs, but not systematically in the wrong direction. This model's R^2 is lower still (0.024): these covariates identify where errors concentrate, not most of the residue-level variation.

Two pre-specified descriptive analyses addressed possible confounds. First, per-chain AUPR drop showed no meaningful association with chain length (Spearman $\rho = 0.028$, 95% CI [-0.054, 0.108], $p = 0.513$). This contradicts the analysis plan's stated expectation that chain length would explain why the pooled AUPR gap (0.070) exceeds the per-chain median gap (0.027). The drop distribution is instead right-skewed (unweighted mean 0.061 vs. median 0.027), and weighting the mean by chain length barely moves it (0.061). A minority of chains with unusually large drops pull the mean above the median; long chains in general do not. Two post-hoc checks (Section 2.4) apportion the rest of the pooled-vs.-median difference. Removing the 55 chains (10%) with the largest per-chain drops roughly halves the pooled gap, from 0.070 (CI [0.051, 0.091]) to 0.039 (CI [0.024, 0.054]). That remainder still exceeds the remaining chains' mean per-chain drop (0.013). Replacing each chain's scores by their within-chain ranks, which removes differences in score calibration between chains, leaves a pooled gap of 0.055 (CI [0.039, 0.072]). Rank-based pooled AUPR is on a different scale from the raw one, so this does not measure a calibration share. It does show that the pooled gap is not mainly a calibration artifact: it persists when only within-chain rankings are compared. The large pooled gap is thus driven substantially, but not entirely, by the right tail of per-chain drops. (Removing chains selected on their drop is a diagnostic, not an unbiased estimate.) Second, the 92 geometrically flagged chains differ from the remaining 462 on every measure in Table 4 except interface fraction. Their larger global Cα distances follow from how the flag is defined, and their larger local RMSD ($p < 10^{-10}$) is closely related. The evidence that does not depend on the flag's definition is AlphaFold's own confidence: flagged chains have substantially lower mean pLDDT ($p < 10^{-10}$), which is consistent with genuine

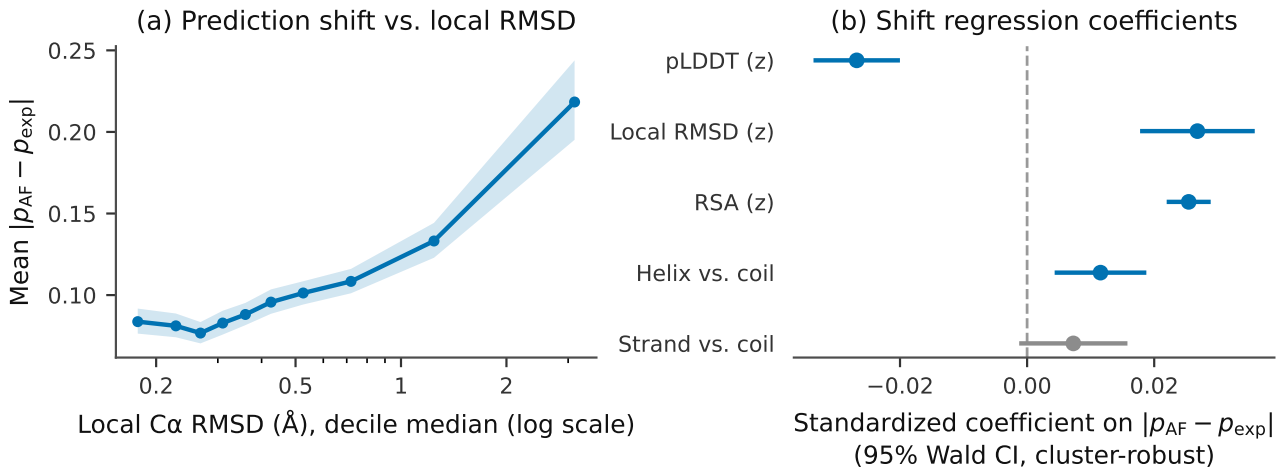


Figure 5: (a) Mean absolute prediction shift in each decile of local Cα RMSD (equal numbers of residues per decile), plotted at the decile's median local RMSD on a log scale, with chain-resampled 95% CIs. (b) Standardized coefficients (95% Wald CI, cluster-robust) from the per-residue shift regression; grey markers did not reach $p < 0.05$.

Term	Coef.	Cluster-robust SE	t	p	VIF
Intercept	0.1019	0.0029	35.17	$< 10^{-10}$	–
pLDDT (z)	-0.0268	0.0033	-8.14	$< 10^{-10}$	1.64
Local RMSD (z)	0.0268	0.0044	6.05	1.4×10^{-9}	1.48
RSA (z)	0.0254	0.0016	15.99	$< 10^{-10}$	1.18
Helix vs. coil	0.0115	0.0035	3.30	9.8×10^{-4}	1.15
Strand vs. coil	0.0073	0.0042	1.74	0.081	1.18

Table 2: Regression of $|p_{AF} - p_{exp}|$ on standardized predictors, cluster-robust standard errors by chain. Reference category: secondary structure = coil.

AlphaFold-vs-experimental divergence rather than a mapping artifact, though it does not rule out the latter for individual chains. Flagged chains are also longer, consistent with (but not evidence for) domain rearrangement in multi-domain chains. A post-hoc audit of their residue mappings found them only slightly less clean than the unflagged chains. 7 of 92 flagged chains used the fallback alignment, versus 19 of 462 unflagged chains; 24 versus 73 had at least one sequence mismatch. Mean mapped identity was 99.6% versus 99.8% (Mann–Whitney $p = 0.024$). Mapping problems may contribute for a few flagged chains but do not characterize the group.

Figure 6 illustrates these results on a single chain (PDB 7DZ9, chain C, UniProt E3BK13, 180 residues), chosen because its AUPR drop (0.031) sits close to the primary endpoint's median. PeSTo's experimental-input predictions (AUPR 0.837) track the true interface patch more tightly than its AlphaFold-input predictions (AUPR 0.806) on the same residues, though both visibly recover the same overall interface region.

3.3 Does pLDDT filtering actually help? (post-hoc, exploratory)

Section 3.2's regression shows pLDDT is *associated* with the size of the prediction shift, but that does not by itself establish that discarding low-pLDDT residues is a good deployment rule. This subsection tests that directly. It was not part of either pre-specified plan: the question arose only after the results of Sections 3.1 and 3.2 were known. Its specification (the cutoffs, the fixed probability threshold, the experimental-input reference curve, and the requirement to report the cost) was written down before these specific numbers were computed. As with the other plans, that ordering is documented only in the author's records, and the analysis was committed together with its results (`ec4b3d1`). It is exploratory and is not confirmatory evidence.

For both inputs, at a fixed probability threshold (0.5, not tuned per input or cutoff) we swept a pLDDT cutoff (0, 50, 70, 90) that discards residues below it, and recomputed precision, recall, and F1 on what remains (Table 5, Fig. 7). The identical sweep on *experimental* input serves as a reference. Since pLDDT is a property of the AlphaFold model shared by both inputs' rows, any F1 change on the *exp* curve reflects only the generic effect of discarding residues, not anything specific to AlphaFold's structural error; comparing the two curves isolates the latter.

	Shift $\lvert p_{\mathrm{AF}} - p_{\mathrm{exp}} \rvert$			Signed error $\lvert p_{\mathrm{AF}} - y \rvert - \lvert p_{\mathrm{exp}} - y \rvert$		
Term	Coef.	t	p	Coef.	t	p
Intercept	0.1019	35.2	$< 10^{-10}$	0.0160	7.0	$< 10^{-10}$
pLDDT (z)	-0.0268	-8.1	$< 10^{-10}$	-0.0145	-4.0	7.1×10^{-5}
Local RMSD (z)	0.0268	6.1	1.4×10^{-9}	0.0233	3.9	8.2×10^{-5}
RSA (z)	0.0254	16.0	$< 10^{-10}$	-0.0012	-0.7	0.473
Helix vs. coil	0.0115	3.3	9.8×10^{-4}	0.0048	1.4	0.161
Strand vs. coil	0.0073	1.7	0.081	0.0036	0.9	0.363
R^2	0.093			0.024		

Table 3: Post-hoc. The regression design of Table 2 (same 95,117 residues, cluster-robust standard errors by chain) with the pre-specified outcome (left) and the signed error (right; positive when the AlphaFold-input prediction is further from the true label y).

Metric	Flagged mean (median)	Unflagged mean (median)	p
Per-chain AUPR drop	0.108 (0.059)	0.052 (0.024)	0.020
Chain length (residues)	224 (202)	162 (117)	2.5×10^{-4}
Interface fraction	0.314 (0.215)	0.327 (0.266)	0.179
Mean pLDDT	77.4 (84.1)	87.9 (90.6)	$< 10^{-10}$
Mean local RMSD (Å)	2.21 (1.39)	0.67 (0.55)	$< 10^{-10}$
Mean global Cα dist. (Å)	7.91 (7.37)	1.40 (0.98)	$< 10^{-10}$

Table 4: Geometrically flagged vs. unflagged primary-set chains (two-sided Mann–Whitney U test).

Filtering at 90 discards 34% of all residues and, importantly, 43% of the true interface residues, while moving AlphaFold-input F1 from 0.596 (CI [0.571, 0.621]) to 0.603 (CI [0.573, 0.633]), an absolute gain of 0.007. Because the two F1 values are computed on nested, correlated residue sets, their marginal CIs cannot be compared directly; a post-hoc paired bootstrap (both cutoffs evaluated on the same resampled chains) gives a 95% CI of [-0.011, 0.024] for this gain, which includes zero. The experimental-vs.-AlphaFold F1 gap does narrow, from 0.050 without filtering to 0.031 at the strictest cutoff (a change of -0.019, paired CI [-0.031, -0.008], i.e. 39% of the original gap). About 64% of that narrowing comes from experimental-input F1 falling by 0.013 (0.647 to 0.634; paired CI for the change [-0.028, 0.003]) as its harder, lower-pLDDT residues are discarded too, rather than from AlphaFold input improving. Filtering thus partly closes the gap between the inputs, but it does not detectably improve AlphaFold-input performance itself. The composition of the small gain is also informative: AlphaFold-input precision rises from 0.673 to 0.713 while recall is essentially flat (0.535 to 0.523). Filtering trades a large, real cost in discarded true interface residues for a modest precision gain, not a broad quality improvement. At a fixed 0.5 threshold, its practical benefit here is limited relative to its cost.

4 Discussion

In practical terms, PeSTo's accuracy cost from using an AlphaFold structure instead of an experimental one is real, directionally consistent, and statistically unambiguous, but modest for a typical chain: a median AUPR difference around 0.027 on a 0–1 scale is a small-to-moderate effect, not a collapse in accuracy, and Fig. 2a shows most chains clustering close to the identity line, with AlphaFold input actually scoring higher for 32% of chains. The pooled-residue comparison (0.739 vs. 0.669) suggests a larger gap. Post-hoc checks attribute much of that difference to a right-skewed minority of chains with large drops: removing the 10% with the largest drops roughly halves it. The remainder reflects pooled AUPR ranking residues from all chains together, rather than averaging per-chain values (Section 3.2). Reporting the per-chain median alongside the pooled number avoids overstating the typical case.

This study's numbers are not directly comparable to Yuan et al.'s ESMFold-based benchmark (median per-protein AUPR 0.797 to 0.691) [8]: the structure predictor, the test set, the leakage control, and the interface definition all differ, and only the direction of the effect – a real accuracy cost from using a predicted structure – is expected to generalize between the two studies, not the magnitude. AlphaFold's average accuracy is generally reported as higher than ESMFold's [9], which is at least directionally consistent with this study's smaller median per-chain gap, but the two benchmarks differ in enough other respects that this is a plausible reading of the comparison, not a controlled one.

The error analysis in Section 3.2 indicates where the gap concentrates, rather than a diffuse "AlphaFold structures

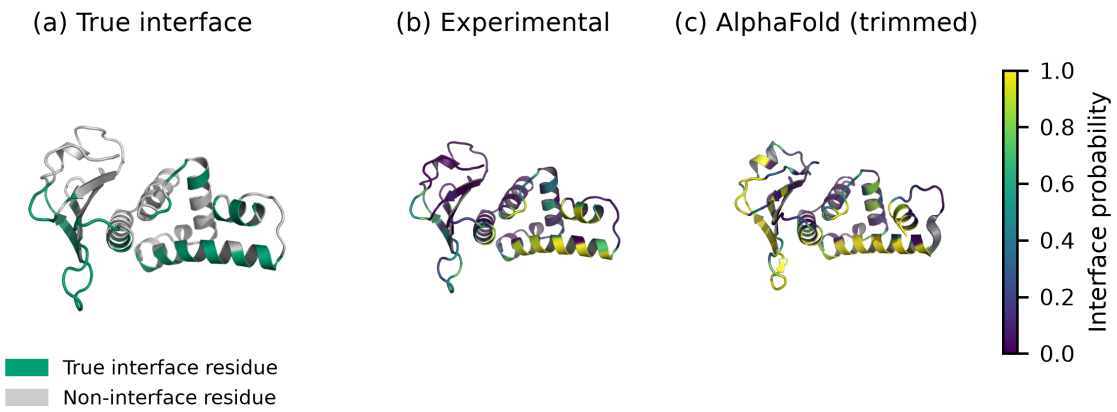


Figure 6: A representative chain (rendered with PyMOL (cartoon representation), not a Cα trace; the AlphaFold model in (c) is Kabsch-superposed onto the experimental structure and all three panels share one camera orientation; produced by scripts/render_example_chain.py, see Code and Data Availability). (a) True interface residues (green) vs. non-interface (grey). (b) PeSTo's per-residue interface probability from the experimental structure (AUPR 0.837 on this chain). (c) PeSTo's per-residue interface probability from the AlphaFold model trimmed to the same residue range (AUPR 0.806, scored on the same residues as (b)), superposed onto the experimental frame for comparison only (the underlying coordinates, and hence the prediction, come from the AlphaFold model).

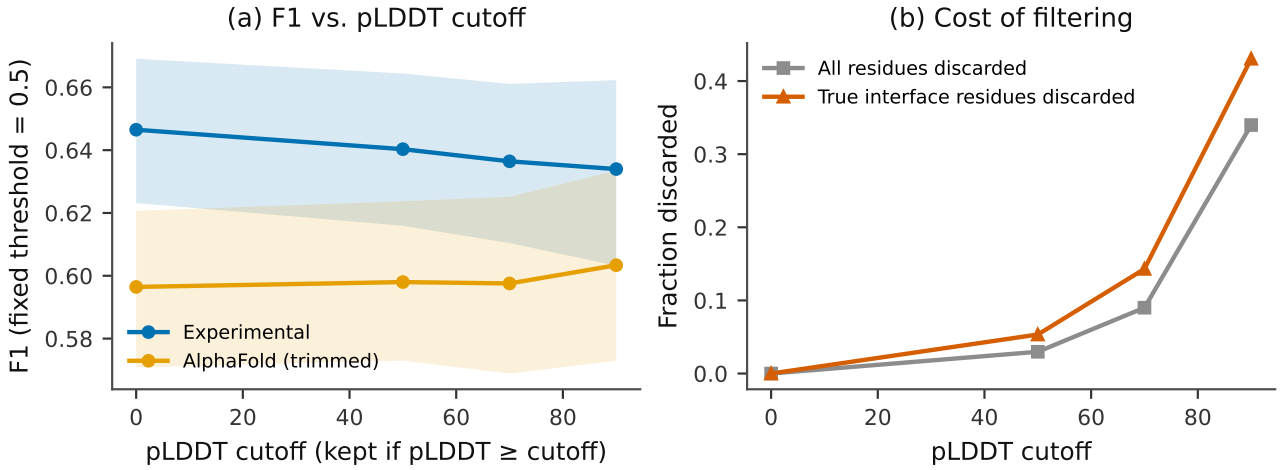


Figure 7: (a) F1 at a fixed 0.5 probability threshold vs. pLDDT cutoff, both inputs, with 95% chain-resampled bootstrap CIs. (b) Fraction of all residues, and of true interface residues specifically, discarded at each cutoff (identical for both inputs, since pLDDT is a property of the AlphaFold model, not of which input is scored).

are worse" pattern: the per-residue cost is largest where AlphaFold itself reports low confidence and where the local backbone geometry diverges from the experimental structure, and these two signals carry partly independent information (low collinearity despite both tracking the same underlying divergence). A post-hoc refit on the signed error shows that both signals also mark where AlphaFold-input predictions move *away* from the true label, not merely where they change. Solvent exposure is the strongest correlate of how much a prediction changes, which is unsurprising for a method whose interface calls are made on the protein surface, but it is not associated with the direction of the error. These are associations, and the covariates explain only a small fraction of residue-level variation. Association with a continuous shift is also not the same as a useful deployment rule. Section 3.3's direct test of the natural follow-on idea – discard low-pLDDT residues before trusting AlphaFold-input predictions – found limited practical benefit at a fixed threshold: AlphaFold-input F1 changed by only 0.007 from no filtering to the strictest cutoff, while 43% of the true interface residues were discarded. pLDDT is a real, independent correlate of where predictions shift the most; that is a weaker and more qualified claim than a validated confidence-based filter, and this paper does not claim the latter.

Input	Cutoff	% res. kept	% interface kept	Precision (95% CI)	Recall (95% CI)	F1 (95% CI)
Experimental	0	100.0	100.0	0.73 [0.71, 0.75]	0.58 [0.55, 0.61]	0.65 [0.62, 0.67]
AlphaFold trimmed		100.0	100.0	0.67 [0.65, 0.70]	0.54 [0.50, 0.57]	0.60 [0.57, 0.62]
Experimental	50	97.0	94.7	0.73 [0.70, 0.75]	0.57 [0.54, 0.60]	0.64 [0.62, 0.66]
AlphaFold trimmed		97.0	94.7	0.69 [0.66, 0.71]	0.53 [0.50, 0.56]	0.60 [0.57, 0.62]
Experimental	70	91.0	85.7	0.73 [0.70, 0.75]	0.57 [0.54, 0.60]	0.64 [0.61, 0.66]
AlphaFold trimmed		91.0	85.7	0.69 [0.67, 0.72]	0.52 [0.49, 0.56]	0.60 [0.57, 0.63]
Experimental	90	66.0	57.0	0.74 [0.71, 0.77]	0.56 [0.52, 0.59]	0.63 [0.60, 0.66]
AlphaFold trimmed		66.0	57.0	0.71 [0.68, 0.74]	0.52 [0.49, 0.56]	0.60 [0.57, 0.63]

Table 5: Precision/recall/F1 (fixed 0.5 threshold) by pLDDT cutoff, both inputs, with 95% chain-resampled bootstrap CIs. Post-hoc, exploratory (Section 3.3).

5 Limitations

- **Template and MSA information in the AlphaFold models.** The release-date filter guarantees that the experimental structures postdate AlphaFold2's training data, not that they postdate the structural templates available when the AlphaFold DB models were generated (all models here are version v6, created 2022-06-01 to 2025-08-01 according to the API metadata). An experimental structure, or a close homolog, released before its model was generated could have served as a template for that model. That would make the AlphaFold input more similar to the experimental structure and bias the measured gap toward zero. The exact template cutoff is not recorded in the cached metadata. The post-hoc check in Section 3.1 shows the gap is present among the 232 chains released after their model was generated, which rules out self-templating for that subgroup, but not templating on earlier close homologs.
- **Temporal and compositional skew.** Leakage filtering removed 77% of the non-redundant representatives. The remaining set is, by construction, made of proteins dissimilar to PeSTo's training data, and results may not transfer to well-studied families. It is also strongly skewed toward chains released in 2021 or later (release dates 2019-01-16 to 2026-09-16; Supplementary Table S2): almost no representative released in 2018–2020 survives the filter, which suggests PeSTo's training-data snapshot extends to roughly 2020–2021, later than AlphaFold2's cutoff.
- **Leakage definition.** Leakage was assessed by sequence identity alone, and a hit counted only if its alignment covered at least 80% of both sequences. Homology confined to one domain of a multi-domain chain, or structural similarity without detectable sequence identity, is therefore not excluded.
- **Dataset scope.** Only X-ray structures at or better than 2.5 Å were used. 97 chains differ from their AlphaFold model's sequence at one or more mapped positions (identity $\geq$90% was allowed, to retain engineered constructs), so for these the two inputs differ slightly in sequence as well as structure. The 40-residue length filter applies to the expressed entity, not to the resolved coordinates. Some analyzed chains are therefore short resolved fragments, including the 12 chains that are entirely interface.
- **Bound vs. unbound conformations.** All experimental structures here are bound (holo) structures taken from complexes; a protein's shape can change somewhat upon binding, and this study cannot separate that ordinary conformational change from AlphaFold-specific structural error. A planned comparison against independently solved unbound (apo) structures was not run.
- **Shift rather than error.** The pre-specified regression outcome $|p_{\mathrm{AF}} - p_{\mathrm{exp}}|$ measures how much a residue's prediction changes between inputs, not whether the AlphaFold-input prediction is worse. The signed-error refit that addresses this is post-hoc, and both models have low R^2.
- **Deferred comparisons.** The untrimmed, full-length AlphaFold model (as opposed to the same model trimmed to the experimentally observed range) and the leakage-sensitivity subsets at looser identity thresholds were not run; the trimmed-model result reported here should not be read as characterizing performance on a full-length AlphaFold model used without any trimming.
- **Single predictor, single confidence signal.** All results are specific to PeSTo; whether they generalize to ScanNet [6] or other residue-level interface predictors such as GPSite [8] or MaSIF-site [19] is untested here. (HIGH-PPI [11] primarily predicts whether two proteins interact, so it is a less direct comparator.) Only pLDDT was used as AlphaFold's confidence signal; AlphaFold DB also provides predicted aligned error (PAE) for these monomer models, but PAE was outside this study's scope.
- **Monomeric AlphaFold models.** Every AlphaFold structure used here is a single-chain model; no multi-chain (complex) AlphaFold prediction was evaluated, so this study speaks to isolated-chain structure quality, not to whether an AlphaFold-predicted complex's own interface would look different.

6 Future Work

The deferred comparisons noted above – the untrimmed full-length AlphaFold model, the unbound (apo) structures, and the leakage-sensitivity subsets at looser identity thresholds – remain natural next steps and require no new method, only additional runs of the existing pipeline. Section 3.3 found that simply *discarding* low-pLDDT residues at a fixed threshold recovers little of the measured accuracy cost, which is itself informative: it suggests the useful information in pLDDT is more continuous than a hard cutoff can exploit. A pLDDT-*aware* fine-tuning of PeSTo – adding pLDDT as an extra per-residue input channel, training on a mixed set of experimental structures (pLDDT fixed at 100) and AlphaFold structures (their real pLDDT) – remains a plausible next step, since a trained model could in principle learn a more graded use of the signal than any fixed cutoff; but Section 3.3 means this should be treated as an open question to test, not a predicted win.

Code and Data Availability

All code, the analysis plans, and the scripts that regenerate every number, table, and figure in this paper from the pipeline's outputs are available at `https://github.com/Menrae/ppi-interface-transfer`. Supplementary Table S1, listing every chain in the primary benchmark set with its UniProt accession, assembly, mapping method, flag status, per-chain AUPRs, and whether it enters the per-chain analyses, is `paper/supplementary/table_s1_primary_set_chains.csv` in that repository. The structure rendering in Fig. 6 was produced with PyMOL (open-source build, version 3.1.0) [20] by `scripts/render_example_chain.py`. Input data are public: PDB entries via PDBe, AlphaFold DB models via the AlphaFold DB API, and PeSTo's model and dataset splits via its GitHub repository.

AI Assistance Statement

This project made substantial use of AI assistance, and we describe it here so that readers can judge the work accordingly. The research question, the original project proposal, and the decisions that shaped the study (scope, leakage policy, which analyses to run, and how to respond to review) are the author's. Claude Code, an AI coding assistant developed by Anthropic and running Claude large language models, was used throughout, under the author's direction, for:

- writing the data pipeline, analysis code, and tests in the accompanying repository, and running them;
- drafting the written analysis plans and this manuscript's text, figures, and tables, which the author reviewed and revised;
- checking the manuscript against the pipeline's outputs, and checking quoted literature values and citation metadata against the original sources and Crossref.

A separate Claude instance, given only the compiled PDF, also produced a critical review of an earlier draft. Its points were checked against the repository before being accepted, disputed, or acted on. The post-hoc analyses in this paper (Section 2.4) originated from that review.

Several safeguards limit the risk of AI-introduced errors. Every number in the paper is generated by a script from the pipeline's own outputs rather than typed by hand. All analysis code has automated tests. The analysis plans and full version history are public in the repository. The AI tool is not listed as an author because it cannot take responsibility for the work. The author takes full responsibility for the content of this paper, including any errors.

Funding

This work received no funding.

References

[1] Zhang, J., Durham, J., Cong, Q. Revolutionizing protein–protein interaction prediction with deep learning. *Curr Opin Struct Biol* **85**, 102775 (2024). `https://doi.org/10.1016/j.sbi.2024.102775`

[2] Jumper, J. et al. Highly accurate protein structure prediction with AlphaFold. *Nature* **596**, 583–589 (2021). `https://doi.org/10.1038/s41586-021-03819-2`

[3] Varadi, M. et al. AlphaFold Protein Structure Database in 2024: providing structure coverage for over 214 million protein sequences. *Nucleic Acids Res* **52**, D368–D375 (2024). https://doi.org/10.1093/nar/gkad1011

[4] EMBL-EBI. pLDDT: understanding local confidence. AlphaFold online training course. https://www.ebi.ac.uk/training/online/courses/alphafold/inputs-and-outputs/evaluating-alphafolds-predicted-structures-using-confidence-scores/plddt-understanding-local-confidence/ (accessed 2026-10-01).

[5] Krapp, L.F., Abriata, L.A., Cortés Rodríguez, F. et al. PeSTo: parameter-free geometric deep learning for accurate prediction of protein binding interfaces. *Nat Commun* **14**, 2175 (2023). https://doi.org/10.1038/s41467-023-37701-8

[6] Tubiana, J., Schneidman-Duhovny, D., Wolfson, H.J. ScanNet: an interpretable geometric deep learning model for structure-based protein binding site prediction. *Nat Methods* **19**, 730–739 (2022). https://doi.org/10.1038/s41592-022-01490-7

[7] Terwilliger, T.C. et al. AlphaFold predictions are valuable hypotheses and accelerate but do not replace experimental structure determination. *Nat Methods* **21**, 110–116 (2024). https://doi.org/10.1038/s41592-023-02087-4

[8] Yuan, Q. et al. Genome-scale annotation of protein binding sites via language model and geometric deep learning. *eLife* **13**, RP93695 (2024). https://doi.org/10.7554/eLife.93695

[9] Lin, Z. et al. Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science* **379**, 1123–1130 (2023). https://doi.org/10.1126/science.ade2574

[10] Krissinel, E., Henrick, K. Inference of macromolecular assemblies from crystalline state. *J Mol Biol* **372**, 774–797 (2007). https://doi.org/10.1016/j.jmb.2007.05.022

[11] Gao, Z., Jiang, C., Zhang, J. et al. Hierarchical graph learning for protein–protein interaction. *Nat Commun* **14**, 1093 (2023). https://doi.org/10.1038/s41467-023-36736-1

[12] Steinegger, M., Söding, J. MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nat Biotechnol* **35**, 1026–1028 (2017). https://doi.org/10.1038/nbt.3988

[13] Velankar, S. et al. SIFTS: Structure Integration with Function, Taxonomy and Sequences resource. *Nucleic Acids Res* **41**, D483–D489 (2013). https://doi.org/10.1093/nar/gks1258

[14] Kunzmann, P., Hamacher, K. Biotite: a unifying open source computational biology framework in Python. *BMC Bioinformatics* **19**, 346 (2018). https://doi.org/10.1186/s12859-018-2367-z

[15] Shrake, A., Rupley, J.A. Environment and exposure to solvent of protein atoms. Lysozyme and insulin. *J Mol Biol* **79**, 351–371 (1973). https://doi.org/10.1016/0022-2836(73)90011-9

[16] Tsai, J., Taylor, R., Chothia, C., Gerstein, M. The packing density in proteins: standard radii and volumes. *J Mol Biol* **290**, 253–266 (1999). https://doi.org/10.1006/jmbi.1999.2829

[17] Tien, M.Z., Meyer, A.G., Sydykova, D.K., Spielman, S.J., Wilke, C.O. Maximum allowed solvent accessibilities of residues in proteins. *PLoS ONE* **8**, e80635 (2013). https://doi.org/10.1371/journal.pone.0080635

[18] Labesse, G., Colloc'h, N., Pothier, J., Mornon, J.-P. P-SEA: a new efficient assignment of secondary structure from Cα trace of proteins. *Comput Appl Biosci* **13**, 291–295 (1997). https://doi.org/10.1093/bioinformatics/13.3.291

[19] Gainza, P. et al. Deciphering interaction fingerprints from protein molecular surfaces using geometric deep learning. *Nat Methods* **17**, 184–192 (2020). https://doi.org/10.1038/s41592-019-0666-6

[20] The PyMOL Molecular Graphics System, open-source version. Schrödinger, LLC. https://github.com/schrodinger/pymol-open-source

Appendix: release-year breakdown of the leakage filter

Release year	Representatives	Leakage-free	% flagged	Analyzed (primary set)
2018	320	0	100.0	0
2019	416	1	99.8	1
2020	385	3	99.2	3
2021	328	96	70.7	87
2022	341	128	62.5	107
2023	335	127	62.1	102
2024	333	130	61.0	104
2025	331	122	63.1	90
2026	220	89	59.5	72

Table S2: Non-redundant representatives by PDB release year: total, number below the 30% identity leakage threshold, percentage flagged as leakage, and number in the final 566-chain primary benchmark set.